Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Half Past Paper – Week 14

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
2	Multiplying with mixed numbers	3	U224
4	Congruent triangles	1	U866
6	Angles on parallel lines, Combining angle facts	5	U826,U655
8	Calculating with pressure, Finding the volume of prisms	3	U527,U174
10	Combining ratios	3	U921
12a	Cumulative frequency graphs	1	U182,U642
12b	Cumulative frequency graphs	2	U182,U642
12c	Interpreting cumulative frequency graphs	3	U642
14	Tree diagrams for independent events	3	U558
16a	Selecting sampling techniques	3	U162
16b	Selecting sampling techniques	1	U162
18	Solving direct proportion word problems, Percentage change	3	U721,U671
20	Rationalising denominators containing two terms	4	U281
22	Finding the area of sectors, Expanding brackets, Factorising to solve quadratic equations	5	U373,U768, U960
	Total	40	

Answer ALL questions.

Write your answers in the spaces provided.

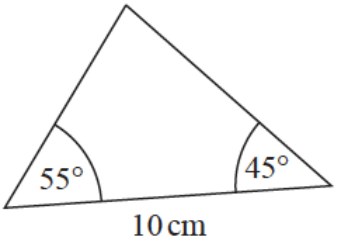
You must write down all the stages in your working.

2 Show that

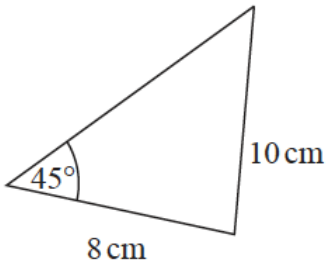
$$2\frac{1}{3}\times 3\frac{3}{4}=8\frac{3}{4}$$

(Total for Question 2 is 3 marks)

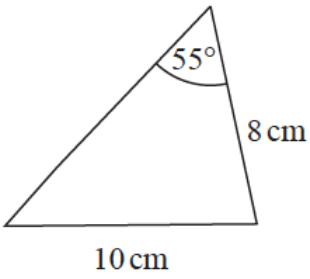
4 The diagram shows four triangles.



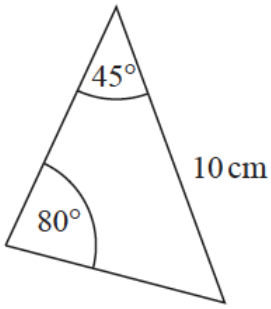
Triangle **A**



Triangle **B**



Triangle **C**



Triangle **D**

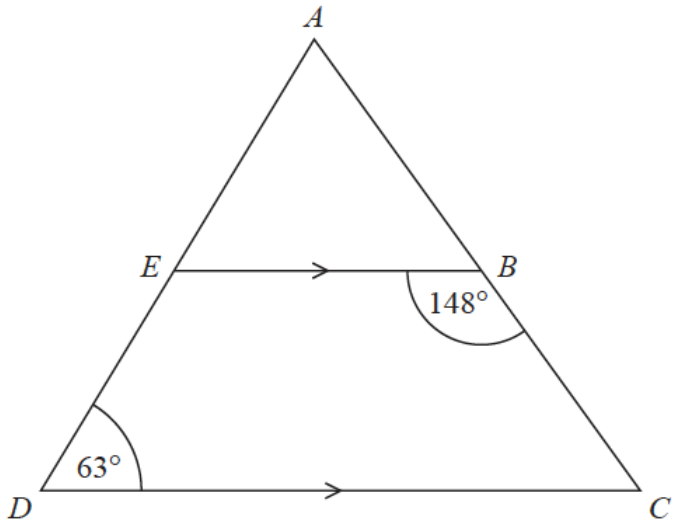
Two of these triangles are congruent.

Write down the letters of these two triangles.

..... and.....

(Total for Question 4 is 1 mark)

6 *ADC* is a triangle.



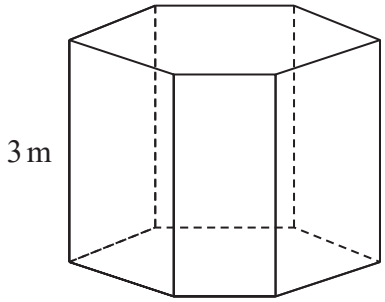
AED and *ABC* are straight lines.
EB is parallel to *DC*.

Angle *EBC* = 148°
Angle *ADC* = 63°

Work out the size of angle *EAB*.
You must give a reason for each stage of your working.

(Total for Question 6 is 5 marks)

8 The diagram shows a prism placed on a horizontal floor.



$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The prism has height 3 m
The volume of the prism is 18 m^3

The pressure on the floor due to the prism is 75 newtons/m^2

Work out the force exerted by the prism on the floor.

..... newtons

(Total for Question 8 is 3 marks)

10 Given that $\frac{a}{b} = \frac{2}{5}$ and $\frac{b}{c} = \frac{3}{4}$

find $a : b : c$

.....

(Total for Question 10 is 3 marks)

12 The table gives information about the weekly wages of 80 people.

Wage (£w)	**Frequency**
$200 < w \leqslant 250$	5
$250 < w \leqslant 300$	10
$300 < w \leqslant 350$	20
$350 < w \leqslant 400$	20
$400 < w \leqslant 450$	15
$450 < w \leqslant 500$	10

(a) Complete the cumulative frequency table.

Wage (£w)	**Cumulative frequency**
$200 < w \leqslant 250$	
$200 < w \leqslant 300$	
$200 < w \leqslant 350$	
$200 < w \leqslant 400$	
$200 < w \leqslant 450$	
$200 < w \leqslant 500$	

(1)

(b) On the grid opposite, draw a cumulative frequency graph for your completed table.

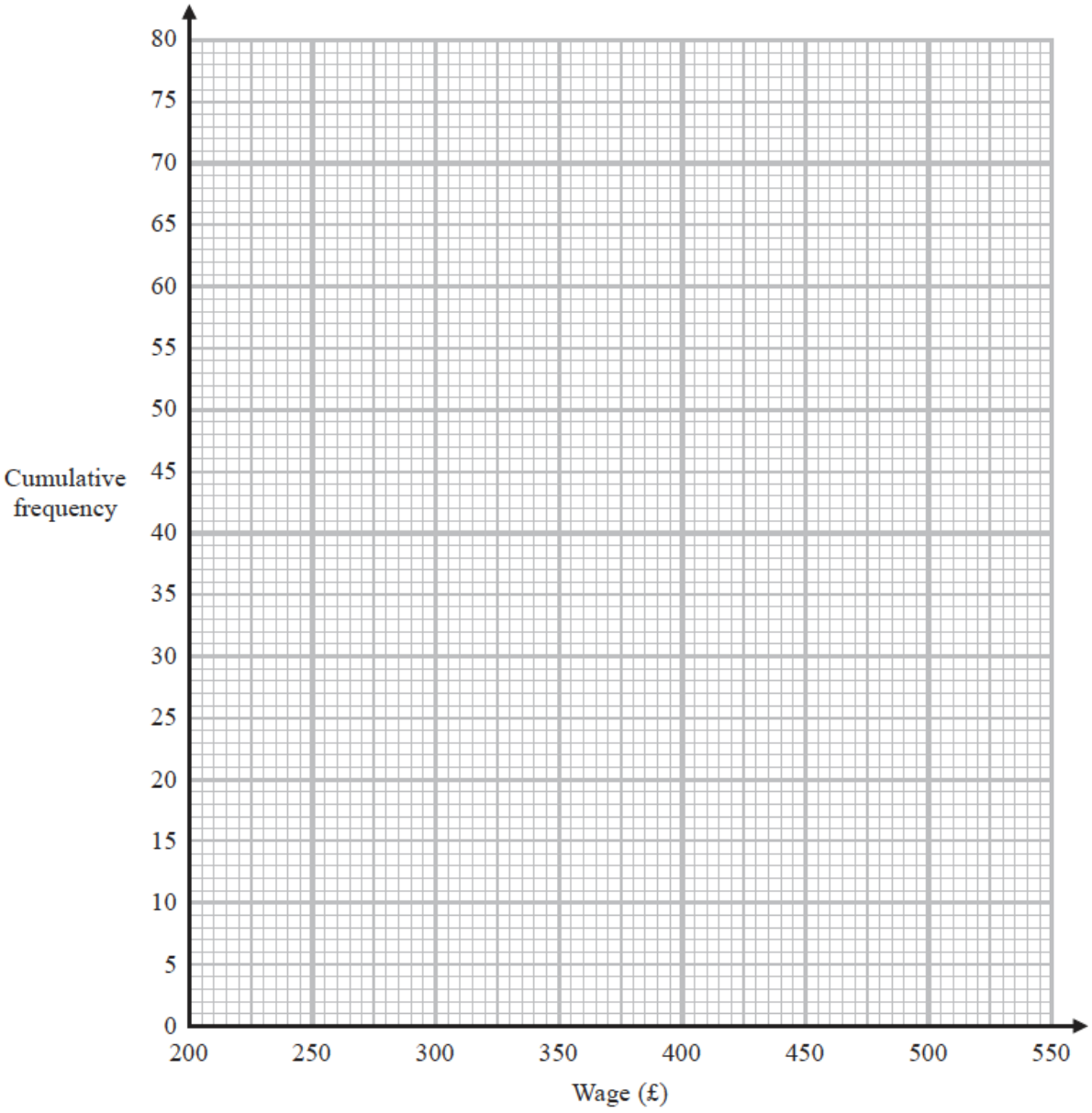
(2)

Juan says

"60% of this group of people have a weekly wage of £360 or less."

(c) Is Juan correct?
You must show how you get your answer.

(3)



(Total for Question 12 is 6 marks)

14 Sally plays two games against Martin.
In each game, Sally could win, draw or lose.

In each game they play,
the probability that Sally will win against Martin is 0.3
the probability that Sally will draw against Martin is 0.1

Work out the probability that Sally will win **exactly** one of the two games against Martin.

.....

(Total for Question 14 is 3 marks)

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16 Shirley wants to find an estimate for the number of bees in her hive.

On Monday she catches 90 of the bees.
She puts a mark on each bee and returns them to her hive.

On Tuesday she catches 120 of the bees.
She finds that 20 of these bees have been marked.

(*a*) Work out an estimate for the total number of bees in her hive.

.....

(3)

Shirley assumes that none of the marks had rubbed off between Monday and Tuesday.

(*b*) If Shirley's assumption is wrong, explain what effect this would have on your answer to part (*a*).

.....

.....

.....

(1)

(Total for Question 16 is 4 marks)

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18 x is proportional to $\sqrt{y}$ where $y > 0$

y is increased by 44%

Work out the percentage increase in x.

.....%

(Total for Question 18 is 3 marks)

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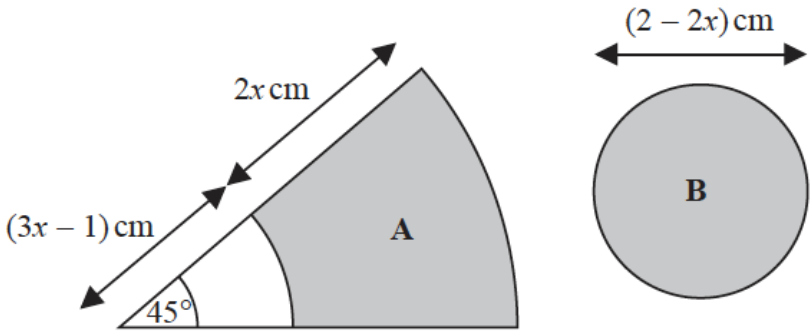
20 Show that $\dfrac{\sqrt{180}-2\sqrt{5}}{5\sqrt{5}-5}$ can be written in the form $a + \dfrac{\sqrt{5}}{b}$ where a and b are integers.

(Total for Question 20 is 4 marks)

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22 The diagram shows two shaded shapes, **A** and **B**.

Shape **A** is formed by removing a sector of a circle with radius $(3x - 1)$ cm from a sector of the circle with radius $(5x - 1)$ cm.
Shape **B** is a circle of diameter $(2 - 2x)$ cm.



The area of shape **A** is equal to the area of shape **B**.

Find the value of x.
You must show all your working.

.....

(Total for Question 22 is 5 marks)

TOTAL FOR PAPER IS 40 MARKS